L05 - Electrocardiography (ECG) I (4.1.7 Nov 3 2022)
File name: Team5a-L05
4/2/2026
10:01 AM

First recording (starts at event labeled 'Supine'): Supine and relaxed.

Second recording (starts at event labeled 'Seated'): Seated and relaxed.

Third recording (starts at event labeled 'Deep breathing'): Seated and relaxed for 5 deep breath cycles.

Fourth recording (starts at event labeled 'After exercise'): Seated and relaxed, recovering from exercise.

L05 DATA REPORT

Student's Name: Georgios Karyos (the test subject)

Lab Section: Section A

Date: 17/04/26

I. Data and Calculations

Subject Profile

Name: Georgios Karyos Height:1.79 Weight:79 Age:20 Gender:Male

Note: This Data Report assumes that all lesson recordings were performed, which may not be the case for your lab. Please disregard any references to excluded recordings.

A. Heart Rate

Complete the following tables with the lesson data indicated, and calculate the Mean as appropriate;

Table 5.2

Condition	Cardiac Cycle 40 Value			Mean (calculate)	If CH 40 was not recorded, use 1 BPM
	1	2	3		
Supine	59.289	60.4223	63.6268	61.112	
Seated	63.6268	63.823	62.565	63.338	
Start of inhale	75.949	67.340	67.796	70.361	

Start of exhale	54.495	68.104	72.815	65.138	
After exercise	101.867	93.603	86.206	93.892	

B: Ventricular Systole and Diastole

Table 5.3

Condition	Duration (ms) 1 Delta T	
	Ventricular Systole	Ventricular Diastole
Supine	0.201	0.367
After exercise	0.201	0.265

C: Components of the ECG

Table 5.4

Condition: Supine Recording (Duration measurements taken from 3 cardiac cycles)					
ECG Component	Normative Value Based on resting heart rate 75 BPM	Duration (ms) 1 Delta T			
		1	2	3	Mean (calc)
Waves	Duration (seconds)				
P	.07 - .18	0.188	0.201	0.207	0.198
QRS Complex	.06 - .12	0.093	0.095	0.086	0.091
T	.10 - .25	0.348	0.349	0.346	0.347
Intervals	Duration (seconds)				
P-R	.12 - .20	0.151	0.151	0.154	0.152
Q-T	.32 - .36	0.314	0.308	0.302	0.308
R-R	.80	0.922	0.996	0.999	0.972
Segments	Duration (seconds)				
P-R	.02 - .10	0.219	0.222	0.228	0.223
S-T	< .20	0.367	0.370	0.376	0.371
T-P	0 - .40	0.771	0.854	0.845	0.823

Table 5.4a

Condition: Supine Recording (Amplitude measurements taken from 3 cardiac cycles)					
ECG Component	Normative Value Based on resting heart rate 75 BPM	Amplitude (mV) 1 P-P			
		1	2	3	Mean (calc)

Condition: After Exercise (measurements taken from 1 cardiac cyc			
ECG Component	Normative Values Based on resting heart rate 75 BPM		Duration (ms) 1 Delta T
Waves	Duration (sec)	Amplitude (mV)	
P	.07 - .18	< .20	0.154
QRS Complex	.06 - .12	.10 – 1.5	0.081
T	.10 - .25	< .5	0.198
Intervals	Duration (seconds)		
P-R	.12 - .20		0.131
Q-T	.32 - .36		0.202
R-R	.80		0.589
Segments	Duration (seconds)		
P-R	.02 - .10		0.171
S-T	< .20		0.232
T-P	0 - .40		0.435

Waves	Amplitude (mV)				
P	< 20	0.970	0.950	0.989	0.969
QRS Complex	.10 - 1.5	0.991	1.036	1.065	1.030
T	< 5	0.950	0.989	1.040	0.993

Table 5.5

Note Interpreting ECGs is a skill that requires practice to distinguish between normal variation and those arising from medical conditions. Do not be alarmed if your ECG does not match the “Normative" values and references above and in the Introduction.

OPTIONAL: Paste the annotated ECG component graph (detailed in Step 14 of Data Analysis) below.

II. Questions

D. Using data from table 5.2:

1) Explain the changes in heart rate between conditions. Describe

The patient showed a lower heart rate while lying down (around 59 bpm) and a slightly higher rate when seated (approximately 63 bpm). Heart rate increased at the beginning of inhalation (about 75.95 bpm) and decreased during exhalation (around 54.50 bpm). The highest heart rate was observed after exercise, reaching an average of 93.89 bpm.

2) Are there differences in the cardiac cycle with the respiratory cycle(recording 3 data)?

Yes, differences are present. This is expected, as various factors can influence the measurements, such as slight inaccuracies in sensor placement or imperfect contact with the body.

E. Using data from table 5.3:

What changes occurred in the duration of systole and diastole between resting and post exercise?

The duration of systole remained unchanged between rest and post-exercise (0.201 ms). In contrast, diastole duration decreased after exercise, being longer at rest (0.367 ms) and shorter following exercise (0.265 ms).

F. Using data from tables 5.4 and 5.5:

1) Compared to the resting state, do the durations of the ECG intervals andsegments decrease during exercise? Explain

-Yes, the durations of ECG intervals and segments decreased after exercise compared to rest. This occurs because the body requires more oxygen after physical activity, leading to an increased heart rate, which in turn shortens the ECG intervals.

2) Compare your ECG data to the normative values. Explain any differences.

-Most of the recorded values were close to the expected normal ranges, although some showed slight deviations. These differences may be attributed to natural individual variability or minor measurement inaccuracies.

3) Compare ECG data with other groups in your laboratory. Do their datadiffer? Explain why this may not be unusual.
-The results showed only small variations between groups. This is expected, since heart function differs slightly between individuals, and minor measurement errors or differences in sensor placement can influence the recordings.

G. In order to beat, the heart needs three types of cells. Describe thecells and their function.
1) Pacemaker cells , responsible for initiating electrical signals that trigger each heartbeat.
2) Conducting cells, transmit these electrical impulses throughout the heart to coordinate contraction timing.
3) Contractile cells, generate the force needed to pump blood by contracting.

H. List in proper sequence, starting with the normal pacemaker, elementsof the cardiac conduction system.
1. Sinoatrial (SA) node
2. Atrial conduction pathways
3. Atrioventricular (AV) node
4. Bundle of His
5. Right and left bundle branches
6. Purkinje fibers
7. Activation of ventricular muscle
8. Ventricular contraction, which pumps blood out of the heart

I. I Describe three cardiac effects of increased sympathetic activity, and of increased parasympathetic activity.
Sympathetic:
Increases heart rate
Enhances contractile strength
Speeds up conduction velocity
Parasympathetic:
1) Decreased heart rate
2) Slows conduction through the AV node 3) Slight decrease in atrial contractility

J. In the normal cardiac cycle, the atria contract before the ventricles. Where is this fact represented in the ECG?
In the ECG, atrial contraction occurs after atrial depolarization, which is represented by the P wave.

K. What is meant by "AV delay" and what purpose does the delay serve?
The "AV delay" refers to a short pause in electrical conduction at the atrioventricular node. This delay allows the atria to complete their contraction and fill the ventricles with blood before ventricular contraction begins, ensuring efficient cardiac function.

L. What is the isoelectric line of the ECG?
The isoelectric line is the baseline of the ECG trace, where no overall electrical activity is detected.

M. Which components of the ECG are normally measured along the isoelectric line?
The ECG components typically measured along the isoelectric line are the PR segment and the ST segment.

III. OPTIONAL Active Learning Portion

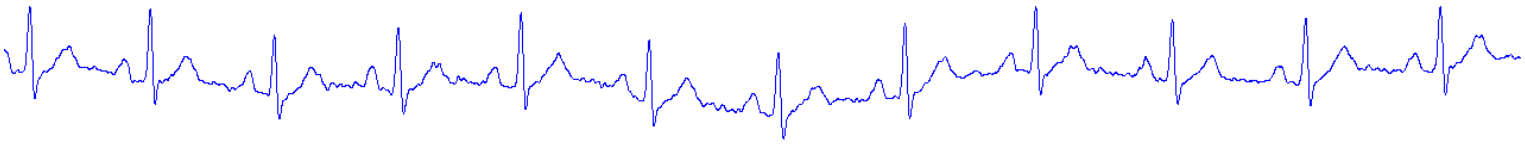
A. Hypothesis

B. Materials

C. Method

D. Set Up

E. Experimental Results



Explain what happens here:

This ECG trace shows a brief irregularity in the heart rhythm. Variations like this can be caused by factors like increased blood pressure due to physical activity or stress, in this case, the irregular pattern is likely because the measurements were taken after exercise, when the body was still recovering. This recovery phase can lead to slight fluctuations in the heart rhythm, explaining the minor inconsistencies observed.

End of Lesson 5 Data Report
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